

This exercise sheet provides some training material for the optional tutorial.

1 Proof patterns

Exercise 1

This is Exercise 1.10 in M. Joshi: Proof Patterns, Springer 2015 [2]. A statement $\mathcal{P}(k)$ is defined for all $k \in \mathbb{Z}$. We know that $\mathcal{P}(\ell)$ is true for some $\ell \in \mathbb{Z}$. Moreover, we know that for all $n \in \mathbb{Z}$,

$$\mathcal{P}(n) \rightarrow \mathcal{P}(n+1) \quad \text{as well as} \quad \mathcal{P}(n) \rightarrow \mathcal{P}(n-1).$$

is true. Prove that $\mathcal{P}(n)$ holds for all $n \in \mathbb{Z}$.

Exercise 2

What is wrong with the following proof of the statement: *for any positive real x and any natural number n , $x^n = 1$ holds?*

Let $\mathcal{P}(n)$ denote $x^n = 1$. The proof is by mathematical induction on n .

Base case. For $n = 0$, $\mathcal{P}(n)$ is true because $x^0 = 1$.

Induction hypothesis. Assume $\mathcal{P}(n)$ is true for some $n \geq 0$.

Induction step. We want to show that $\mathcal{P}(n+1)$ is true. We derive:

$$\begin{aligned} x^{n+1} &= \frac{x^n \cdot x^n}{x^{n-1}} \\ &= \frac{1 \cdot 1}{1} \quad (\text{from above by the induction hypothesis}) \\ &= 1. \end{aligned}$$

Hence, $\mathcal{P}(n+1)$ is true. □

2 Data structures

Exercise 3

Prove the following: *If a tree has $n \geq 1$ vertices, then it has $n - 1$ edges.*

Exercise 4

We define the set L of all lists as follows:

$$L ::= nil \mid (c : L)$$

The symbol nil denotes the empty list containing no element. We define the function *append* by $append(nil, y) = y$ and $append((c : x), y) = (c : append(x, y))$. Show that, for all lists ℓ , $append(\ell, nil) = \ell$ holds.

Exercise 5

For sets $A_1, A_2, \dots$ and $B_1, B_2, \dots$, define $B_1 = A_1$ and $B_n = A_n \setminus (\bigcup_{i=1}^{n-1} A_i)$ for $n > 1$.

Prove: $\bigcup_{i=1}^n B_i = \bigcup_{i=1}^n A_i$.

3 Numbers, sums, etc.

Exercise 6

Show that any natural number above 11 can be written as a sum of 4's and 5's.

Exercise 7

Show that, for all positive integers n , $x^n - y^n$ is divisible by $x - y$.

Exercise 8

Show that for any $n \in \mathbb{N}_0$, the number $\ell_n = 4^{n+2} + 5^{2n+1}$ is divisible by 21.

Exercise 9

Recall from former lectures the definition of *Fibonacci numbers*: $F(0) = 0$, $F(1) = 1$, and for all integers $k \geq 0$, $F(k+2) = F(k+1) + F(k)$. Show that, for all integers $n \geq 0$, $F(n) < 2^n$ holds.

Exercise 10

Recall from former lectures the definition of *Fibonacci numbers*: $F(0) = 0$, $F(1) = 1$, and for all integers $k \geq 0$, $F(k+2) = F(k+1) + F(k)$. Show that, for all integers $n \geq 0$, $\sum_{i=0}^{n-1} F(i) < F(n+1)$ holds.

Exercise 11

Recall from former lectures the definition of *Fibonacci numbers*: $F(0) = 0$, $F(1) = 1$, and for all integers $k \geq 0$, $F(k+2) = F(k+1) + F(k)$. Show that, for all integers $n \geq 3$, $\alpha^{n-2} < F(n)$ holds, where $\alpha = (1 + \sqrt{5})/2$.

Remark: α is called the *golden ratio* and it can be proved that $\lim_{n \rightarrow \infty} F_{n+1}/F_n = \alpha$.

Exercise 12

Let $\mathcal{P}(n)$ denote the statement $n! > 2^n$. Show that there is a smallest natural number n_0 such that $\mathcal{P}(n)$ holds for all natural numbers $n \geq n_0$.

Exercise 13

Let us define the *Fermat* numbers $F_n = 2^{(2^n)} + 1$ for all natural numbers $n \geq 0$. Prove that for all $n \geq 1$, $F_n = (F_0 \cdot F_1 \cdots F_{n-1}) + 2$.

Exercise 14

From M. Sipser: Introduction to the theory of computation, third edition.

Let $S(n) = 1 + 2 + \cdots + n$ be the sum of the first n natural numbers and let $C(n) = 1^3 + 2^3 + \cdots + n^3$ be the sum of the first n cubes. Prove the following equalities by induction on n , to arrive at the curious conclusion that $C(n) = (S(n))^2$ for every n .

1. $S(n) = \frac{1}{2}n(n+1)$.
2. $C(n) = \frac{1}{4}(n^4 + 2n^3 + n^2) = \frac{1}{4}n^2(n+1)^2$.

4 Logic and induction

Exercise 15

Let I, I' be interpretations with domains $\mathcal{U}, \mathcal{U}'$. We say that I is *isomorphic* to I' if there exists a bijection $\chi : \mathcal{U} \rightarrow \mathcal{U}'$ such that the following conditions are fulfilled:

1. $\chi(I(c)) = I'(c)$ for every constant symbol c .
2. $\chi(I(f)(p_1, \dots, p_n)) = I'(f)(\chi(p_1), \dots, \chi(p_n))$ for every n -ary function symbol f ($n > 0$) and all $p_1, \dots, p_n \in \mathcal{U}$.

χ is called *isomorphism* between I and I' .

Now let χ be an isomorphism between I and I' . Prove that for all closed (i.e., variable-free) terms t we have $I'(t) = \chi(I(t))$.

Hint: Prove the statement by structural induction on t .

Exercise 16 *Equivalent Replacement Theorem*

Prove the following statement:

Let φ, ψ_1 , and ψ_2 be propositional formulas. If $\psi_1 \equiv \psi_2$, then $\varphi[\psi_1] \equiv \varphi[\psi_2]$.

5 Binomial coefficients and induction

We consider properties of binomial coefficients in the following. Here are some basic facts and definitions about them.

1. We write binomial coefficients in the form $\binom{a}{b}$, where a and b are non-negative integers.
2. For convenience we allow that b is an integer.
3. $\binom{a}{b}$ is calculated as follows: (i) If $a < b$ or $b < 0$ then $\binom{a}{b} = 0$ by definition. Otherwise $\binom{a}{b} = \frac{a!}{b!(a-b)!}$.
4. $\binom{a}{b} = \binom{a}{a-b}$ and $\binom{a}{0} = \binom{a}{a} = 1$.
5. We will often use that $\binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1}$ for any integer n with $n \geq 0$ and integer r with $0 \leq r \leq n$.

Exercise 17

Show for all non-negative integers n that the following holds:

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n.$$

Exercise 18

Show for all positive integers n that the following holds:

$$\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}.$$

Exercise 19

Show for all non-negative integers n and m that the following holds:

$$\sum_{k=0}^n k \binom{m+k}{m} = n \binom{m+n+1}{m+1} - \binom{m+n+1}{m+2}.$$

Exercise 20

Show for all integers $n > 0$ that the alternating sum of the binomial coefficients is zero, i.e., show that

$$\sum_{\ell=0}^n (-1)^\ell \binom{n}{\ell} = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \cdots \pm \binom{n}{n} = 0.$$

What can you say about the sub-sum of all binomial coefficients $\binom{n}{\ell}$ with $\ell \leq n$ and ℓ even and the similar sub-sum with ℓ odd?

Exercise 21

Show for all integers $n \geq 1$ and all $\ell \geq 1$ that

$$\sum_{i=0}^{\ell} \binom{n+i}{i} = \binom{n}{0} + \binom{n+1}{1} + \cdots + \binom{n+\ell}{\ell} = \binom{n+\ell+1}{\ell}.$$

Hint: Use mathematical induction on ℓ .

Exercise 22

Show that, for all non-negative integers m , k , and n , the following holds:

$$\binom{m+n}{k} = \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i}$$

What can you say about the value of $\binom{2n}{n}$?

Hint: Use mathematical induction on m .

Exercise 23

For positive integers m , n , and p , show that the following holds:

$$\binom{m+n}{m+p} = \sum_{k=0}^m \binom{m}{k} \binom{n}{p+k}$$

Exercise 24

Let $a, b \in \mathbb{R}$, $n \in \mathbb{N}$. Then $(a+b)^n = \sum_{i=0}^n \binom{n}{i} a^{n-i} b^i$.

Exercise 25

For all non-negative integers n and all real numbers x with $x > -1$, $(1+x)^n \geq 1+nx$ holds.

6 More applications

Exercise 26

Suppose we have an operation (denoted by $\prime$) which is applied to symbols. The operation $\prime$ satisfies the following three axioms:

$$\begin{aligned} (u+v)' &= u' + v' & (A+') \\ (uv)' &= uv' + u'v & (A\cdot') \\ (cu)' &= cu' & (Ac') \end{aligned}$$

Define $w^{(k)}$ inductively as follows:

B: $w^{(0)} = w$

S: Suppose $w^{(n)}$ is already defined for some natural number $n \geq 0$. Then $w^{(n+1)} = (w^{(n)})'$.

Prove Leibniz's formula $(uv)^{(n)} = \sum_{i=0}^n \binom{n}{i} u^{(i)} v^{(n-i)}$. (Recall that $\binom{n}{i} + \binom{n}{i-1} = \binom{n+1}{i}$.)

Exercise 27

This exercise is taken from Hopcroft, J.E., Motwani, R. and Ullman, J.D.: *Introduction to Automata Theory, Languages, and Computation*, 3rd ed., Pearson, 2006 [1]. It requires some background from automata theory.

A non-deterministic finite automaton A is a quintuple $(Q, \Sigma, \delta, q_0, F)$, where Q is a finite set of states, Σ is a finite set of input symbols, $q_0 \in Q$ is the start state, and $F \subseteq Q$ is the set of final (or accepting) states. The transition function δ takes a state from Q and an input symbol from Σ as arguments and returns a subset of Q .

The set of all strings over Σ is denoted by Σ^* . Let x and y be two strings. Then xy is the concatenation of them. The length of a string w (i.e., the number of symbol occurrences in w) is denoted by $|w|$. The empty string ϵ is of length 0.

We denote by $\hat{\delta}$ the extension of δ to strings.

$$\hat{\delta}(q, w) = \begin{cases} \{q\} & \text{if } w = \epsilon \text{ (the empty string);} \\ \bigcup_{p \in \delta(q, x)} \delta(p, a) & \text{if } w = xa, x \in \Sigma^* \text{ and } a \in \Sigma. \end{cases}$$

The language accepted by A , $L(A)$, is $\{w \mid \hat{\delta}(q_0, w) \cap F \neq \{\}\}$.

Let $A = (\{q_0, q_1, q_2\}, \{0, 1\}, \delta, q_0, \{q_2\})$, where $\delta(q_0, 0) = \{q_0, q_1\}$, $\delta(q_0, 1) = \{q_0\}$, $\delta(q_1, 0) = \{\}$, $\delta(q_1, 1) = \{q_2\}$, and $\delta(q_2, 0) = \delta(q_2, 1) = \{\}$. Show that $L(A) = \{w \mid w \text{ ends in } 01\}$.

Hint. Use mutual induction on the following statements:

1. $\hat{\delta}(q_0, w)$ contains q_0 for every w .
2. $\hat{\delta}(q_0, w)$ contains q_1 if and only if w ends in 0.
3. $\hat{\delta}(q_0, w)$ contains q_2 if and only if w ends in 01.

Exercise 28

Consider a (generalized) chess board of size $2^n \times 2^n$, where one position is cut out. Take an L-tile made of three positions and show for all natural numbers $n \geq 1$ that the chess board can be covered using the L-tiles. Compute the number of required L-tiles.

Exercise 29

You have a bag with red, yellow and blue chips. If only one chip remains in the bag, you put it out. Otherwise you remove two chips at random:

1. If one of the removed chips is red, you do not put any chips in the bag.
2. If both of the removed chips are yellow, you put one yellow chip and five blue chips in the bag.
3. If one of the chips is blue and the other is not red, you put ten red chips in the bag.

Show that any sequence of moves applied to an arbitrary bag always terminates or provide a non-terminating sequence of moves.

Exercise 30

This exercise requires some background from linear algebra, which we take from https://en.wikipedia.org/wiki/Matrix_exponential.

A *matrix exponential* is a matrix function defined as

$$e^M = \sum_{k=0}^{\infty} \frac{1}{k!} M^k = I + M + \frac{M^2}{2!} + \dots$$

where M is an $n \times n$ (real or complex) matrix and $M^0 = I$ is the *identity* matrix. The series e^M *converges absolutely* (see <http://math.ucr.edu/~res/math138A/expmatrix.pdf> for the definition and a proof). Absolute convergence implies that manipulations on infinite sums like the Cauchy product

$$\left(\sum_{i=0}^{\infty} A_i \right) \left(\sum_{j=0}^{\infty} B_j \right) = \left(\sum_{i,j=0}^{\infty} A_i B_j \right) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n A_k B_{n-k} \right)$$

can be safely performed.

Show the following:

1. If $AB = BA$ then $A^\ell e^B = e^B A^\ell$ and $e^A e^B = e^B e^A$ for arbitrary $n \times n$ (real or complex) matrices A and B , and $\ell \in \mathbb{N}_0$.
2. $e^{(s+t)A} = e^{sA} e^{tA}$ for an arbitrary $n \times n$ (real or complex) matrix A and (real or complex) numbers s and t . What can you say about the inverse of e^A ?

Exercise 31

In this exercise, I present a proof of one of Ramanujan's famous equations

$$\sqrt{1 + n\sqrt{1 + (n+1)\sqrt{\dots}}} = n + 1. \quad (\text{Rn})$$

Here and in the following, a square root means the positive square root. The proof follows M. Sheard's proof in [3] but additional explanations are added. There are some claims for which you should provide a proof.

Let us start by defining for all $1 \leq n \leq m$ the expression

$$g(n, m) = \sqrt{1 + n\sqrt{1 + (n+1)\sqrt{\cdots (m-1)\sqrt{1+m}}}}$$

So $g(m, m) = \sqrt{1+m}$ and $g(n, m) = \sqrt{1 + ng(n+1, m)}$ for $1 \leq n < m$.

Claim 1: For all n, m with $1 \leq n \leq m$, we have $g(n, m) \leq n+1$.

Prove Claim 1 by a downward induction on n starting from m and proceeding towards 1.

Next we consider the sequence $g(n, m)$ for a fixed n and increasing $m \geq n$.

Claim 2: For all natural numbers $m \geq n$, $g(n, m) \leq g(n, m+1)$ holds.

Prove Claim 2 in a similar way as Claim 1.

When we consider the sequence, then we see that (i) it is bounded and (ii) it is increasing. Recall from your math lecture the following theorem.

Every increasing bounded sequence converges (i.e., has a limit).

Therefore we can, for any positive integer n , define $f(n)$ as follows.

$$f(n) = \lim_{m \rightarrow \infty} g(n, m) = \sqrt{1 + n\sqrt{1 + (n+1)\sqrt{\cdots}}} \leq n+1$$

We need next that $f(n) \leq f(n+1)$. First we argue that $\forall n f(n) \geq 1$. Suppose there exists an n , for which $f(n) < 1$. Then $f(n) = \sqrt{1 + nf(n+1)} < 1$ which implies that $f(n+1)$ is negative. But this is impossible, because $f(n+1)$ is of the form $\sqrt{1 + (n+1)f(n+2)}$ and we only consider the positive square root.

We argue that $f(n) > n$. Since $g(n, m)$ is an “initial expression” of $f(n)$ and $g(n, m)$ is increasing for increasing m , it suffices to show that $g(n, m) > n$ for a large enough value for m .

Claim 3: Let $m = n^2$. Then for all n , $g(n, n^2) > n$ holds.

Prove Claim 3 in a similar way as Claim 1.

Now we can show that for all positive integers n , $f(n) \leq f(n+1)$ holds. Consider

$$\begin{aligned} f(n)^2 &= 1 + nf(n+1) \\ f(n+1)^2 &= 1 + (n+1)f(n+2) \end{aligned}$$

Now we have

$$\begin{aligned} f(n+1)^2 - f(n)^2 &= 1 + (n+1)f(n+2) - (1 + nf(n+1)) \\ &= (n+1)f(n+2) - nf(n+1) \\ &> (n+1)(n+2) - nf(n+1) \\ &\geq (n+1)(n+2) - n(n+2) \end{aligned}$$

$$= n + 2 > 0.$$

So, $f(n) < f(n + 1)$.

Claim 4: For all n and $k \geq 0$, we have $f(n) > n + 1 - \frac{1}{2^k}$.

Prove Claim 4 by an induction on k .

Note that Claim 4 can be reformulated as follows. First choose an arbitrary n . Then

$$\begin{aligned} \forall k \geq 0 \left(f(n) > n + 1 - \frac{1}{2^k} \right) & \text{ iff } \forall k \geq 0 \left(\frac{1}{2^k} > n + 1 - f(n) \right) \\ & \text{ iff } \forall k \geq 0 \left(|n + 1 - f(n)| < \frac{1}{2^k} \right). \end{aligned}$$

Next we choose an arbitrary $\epsilon > 0$. Instantiate k in such a way that $\frac{1}{2^k} \leq \epsilon$. Then $|n + 1 - f(n)| < \epsilon$ can be derived from $|n + 1 - f(n)| < \frac{1}{2^k}$, $\frac{1}{2^k} \leq \epsilon$ and transitivity. But then we have $\forall \epsilon \geq 0 \ |n + 1 - f(n)| < \epsilon$.

Recall that, for $x, y \in \mathbb{R}$, $x = y$ iff $\forall \epsilon > 0 \ |x - y| < \epsilon$. Therefore, $f(n) = n + 1$.

References

- [1] John E. Hopcroft, Rajeev Motwani, and Jeffrey D. Ullman. *Introduction to Automata Theory, Languages, and Computation*. Addison Wesley, 3 edition, 2006.
- [2] Mark Joshi. *Proof Patterns*. Springer International Publishing, 2015.
- [3] Michael Sheard. Backwards induction and a formula of Ramanujan. *Mathematics Magazine*, 85(5):364–366, 2012.